

Siya Sathaye

<https://github.com/siyasathaye> // Email: siya.sathaye@gmail.com // Phone: 408-712-5726

EDUCATION

University of California, San Diego (UCSD)

B.S. in Data Science

September 2024 – Present | Expected Graduation: 2027

GPA: 3.84/4.0

Relevant Coursework: Linear Algebra, Multivariable Calculus, Data Structures & Algorithms for Data Science, Practical Applications of Data Science (DSC 80)

TECHNICAL SKILLS

Languages: Python, SQL, Java

Data Engineering: ETL pipelines, data cleaning, schema design, time-series & spatiotemporal data

Analytics & ML: Feature engineering, regression, classification, hypothesis testing, similarity search

NLP: Text cleaning, TF-IDF, cosine similarity, language modeling

Visualization & Tools: Pandas, NumPy, Matplotlib, Plotly, scikit-learn, TensorFlow, Git, JupyterHub, FastAPI

PROJECTS & RESEARCH EXPERIENCE

Spatiotemporal Tracking & Trajectory Analysis

Computational Research Intern — Kennedy Lab, Scripps Research | Fall 2025 – Present

- Process and analyze **high-frequency optical-flow tracking data** from jellyfish videos containing **500,000+ frames per recording**.
- Clean, structure, and validate large-scale tracking outputs into **time-series trajectory datasets** using Python (NumPy, Pandas).
- Work with **spatial coordinate data** and **movement dynamics** for downstream behavioral analysis.
- Produce trajectory plots and **animated visualizations of movement over time** to support pattern discovery.

Advanced Data Pipelines & Predictive Modeling

Understanding the Causes of U.S. Power Outages — DSC 80 Final Project

- Built reproducible data pipelines combining climate, infrastructure, and outage event datasets (1,500+ records).
- Performed feature engineering and EDA to analyze event duration and severity patterns.
- Applied permutation-based hypothesis testing and trained a binary classifier predicting outages >24 hours, improving **F1-score from 0.43 to 0.66**.
- Evaluated model fairness across urban vs. rural states.

Language Modeling & Text Mining

- Built unigram and n-gram language models on large text corpora.
- Implemented tokenization, normalization, and probabilistic analysis to extract linguistic patterns.
- Applied **TF-IDF and cosine similarity** techniques for document representation and comparison.

Multimedia Processing & Classification App

- Developed a Python-based image and audio processing application with modular data pipelines.
- Extracted RGB and spatial features from images and trained a KNN classifier for environment classification.
- Emphasized pipeline structure, feature extraction, and model evaluation.

LEADERSHIP & DEVELOPMENT EXPERIENCE

Software Developer — HealthLink (Student Organization)

Board Member | Spring 2025 – Present

- Designing and developing a **full-stack web application from the ground up** to manage member onboarding, authentication, and engagement.
- Implementing a **Supabase-backed membership system**, including user authentication, relational database schema design, and role-based access control.
- Building ETL-style workflows to ingest, validate, and store user signup data, attendance logs, and event participation records.
- Developing backend APIs to query and aggregate membership data for dashboards and administrative views.
- Collaborating on frontend UI/UX to display dynamic member data and activity metrics.

Kalaawishkar Bharatanatyam Dance School — Volunteer Instructor

2019 – 2024

- Taught 100+ hours to students ages 5–10 and coordinated cultural programs with 300+ attendees.

PRODUCT & ENGINEERING EXPERIENCE

Product Intern — Zippy AI

Spring – Summer 2025

- Built a structured Help Center by transforming internal documentation into customer-facing knowledge resources.
- Developed and tested REST APIs using FastAPI and Postman.